# Pre-Analysis Plan: Daughters and Gender Attitudes

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## Introduction and Hypotheses

This study investigates whether the gender of children affects adult attitudes and behavior concerning gender equality and political preferences. Building on prior literature, particularly Oswald and Powdthavee (2010), we examine whether having daughters influences attitudes and behaviors that reflect greater support for gender egalitarian norms and left-wing political preferences.

We test the following hypotheses:

- **H1**: Having daughters increases support for gender equality.
- **H2**: Having daughters increases support for left-wing political preferences.
- **H3**: Having daughters increases men's involvement in housework.
- **H4**: The effect of having daughters on gender attitudes is stronger among men.

## Data and Sample

We use data from the German Socio-Economic Panel (SOEP-Core), focusing on years in which the key dependent variables are available. The sample includes only individuals with non-missing values for the variables used in each estimation. However, we do not drop individuals who have missing values in some waves due to life circumstances -for example, individuals who do not yet have children, or those without income because they are students or homemakers. These respondents remain in the panel and are included in the analysis if the relevant variables are observed.

# Empirical Strategy and Variables

Our aim is to estimate a linear model with two-way fixed effects at the individual level:

$$Y_{it} = \alpha_i + \delta_t + \beta \cdot \text{Daughters}_{it} + \gamma' X_{it} + \varepsilon_{it}$$

Where:

- $\alpha_i$: individual fixed effects
- $\delta_t$: year fixed effects
- $\text{Daughters}_{it}$: a dummy for first-born daughter
- $X_{it}$: vector of time-varying controls

**Controls** The vector of controls $X_{it}$ includes standard socio-demographic characteristics: health status (`ple0008`), education (we use either years of schooling, `pgbilzeit`, or school-leaving degree, `pgpsbil`, depending on availability), and employment status (`plb0022_h`). We also control for marital status (`pld0131_h`) and number of children, which we construct by aggregating observed `kidsex` across waves.

**Robustness** As a robustness check, we may additionally control for gross labor income (`pglabgro`) and risk attitudes (`plh0204_h`), which may capture relevant heterogeneity in socio-economic status and personality traits.

All models include individual fixed effects ($\alpha_i$) and year fixed effects ($\delta_t$). Individuals are included in the panel as long as they have valid observations for the variables of interest in at least one wave. If possible, as a further robustness check, we will restrict the sample to biological parents only, using the `cld0328_h` variable to identify them.

**Identification** To identify causal effects of child gender on attitudes and behaviors, we follow the strategy developed in prior literature (Dahl and Moretti, 2008; Oswald and Powdthavee, 2010). Specifically, we begin by analyzing whether the **first-born child is a daughter**, which is often considered quasi-random and therefore useful for causal inference. As robustness, we may then examine the **total number of daughters** in the household as an alternative treatment intensity measure.

**Outcomes** For gender attitudes, we use a harmonized variable combining `plh0308_v1` and `plh0308_v2`, which were administered to different subsamples within the same wave. We will harmonize variables `plh0308_v1` and `plh0308_v2` by constructing a binary indicator equal to 1 if the average response is greater than or equal to the midpoint of the scale (i.e., closer to "Agree"), and 0 otherwise. This approach allows us to combine the two formats while preserving directional consistency across the waves.

For H2 we will use the variable `plh004`.

For H3 (housework), we use the variable `pli0043_h`, which captures time spent on housework, will be used to construct one categorical outcome variables:

1. A three-level indicator capturing the distribution of housework:
    - 1 = Most of the housework is done by women
    - 2 = Equal distribution of housework
    - 3 = Most of the housework is done by men

To test H4 (heterogeneity by gender), we will:

- Include an interaction term between the daughters variable and respondent gender (`sex_h`)

Standard errors will be clustered at the individual level.

## Summary Table of Variables Used

| Variable | Label (English) | Years Available | Use |
|---|---|---|---|
| plh0308_v1 | Man/Women Equal in Household | 2012 | Outcome (H1, H4) |
| plh0308_v2 | Best if man and woman work the same amount so they can share the responsibility | 2018 | Outcome (H1, H4) |
| plh004 | Political Attitude | 2005;2009;2014;2019 | Outcome (H2) |
| pli0043_h | Housework hours (harmonized) | 2000-2020 | Outcome (H3) |
| kidsex | Child gender | 2000-2020 | Treatment / N. of children |
| gebjahr | Year of birth | 2000-2020 | Construct age variable |
| sex_h | Individual gender | 2000-2020 | Control and interaction for H4 |
| plh0204_h | Risk Attitudes | 2004;2006;2008-2020 | Control |
| ple0008 | Health | 2000-2020 | Control |
| cld0328_h | Biological child | 2010-2020 | Control |
| pld0131_h | Marital Status | 2000-2020 | Control |
| plb0022_h | Employment Status | 2000-2020 | Control |
| pgpsbil | School-Leaving Degree | 2000-2020 | Control |
| pgbilzeit | Years of schooling | 2000-2020 | Control |
| pglabgro | Gross labor income | 2000-2020 | Control |
| syear | Survey year | 2000-2020 | Control |
| netto | Interview status | 2000-2020 | Check data cleaning |
| pid | Person ID | 2000-2020 | Individual fixed effects |
| cid | Household ID | 2000-2020 | Household ID (origin) |
| hid | Household number | 2000-2020 | Household ID |

## Interpretation and Hypothesis Testing

We will evaluate treatment effects based on:

- **Statistical significance**: thresholds at 0.01, 0.05 and 0.10
- **Direction**: consistency with theoretical expectations
- **Magnitude**: effect sizes

## References

Dahl, G. B., & Moretti, E. (2008). The demand for sons. *The Review of Economic Studies*, 75(4), 1085–1120.

Oswald, A. J., & Powdthavee, N. (2010). Daughters and left-wing voting. *The Review of Economics and Statistics*, 92(2), 213–227.